CBSE Test Paper 02

Chapter 4 Motion in A Plane

1. A passenger arriving in a new town wishes to go from the station to a hotel located 10 km away on a straight road from the station. A dishonest cabman takes him along a circuitous path 23 km long and reaches the hotel in 28 min. What is (a) the average speed of the taxi, (b) the magnitude of average velocity? **1**
   a. 47.3 km/hr, 23.4 km/hr
   b. 49.3 km/hr, 21.4 km/hr
   c. 48.3 km/hr, 22.4 km/hr
   d. 46.3 km/hr, 24.4 km/hr
2. A batter hits a baseball so that it leaves the bat at speed v_0 = 37.0 m/s at an angle a = 53.1^o. Find the time when the ball reaches the highest point of its flight, and its height h at this time. **1**
   a. 3.02 s, 44.7 m
   b. 3.32 s, 41.7 m
   c. 3.12 s, 43.7 m
   d. 3.22 s, 42.7 m
3. A unit vector is a vector **1**
   a. having a magnitude of 1 and points in any chosen direction
   b. having a magnitude of 1 and points in x-direction
   c. having a magnitude of 1 and points in y-direction
   d. having a magnitude of 1 and points in z-direction
4. The basic difference between a scalar and vector is one of **1**
   a. magnitude
   b. direction
   c. origin
   d. polar angle

5. A man stands on the roof of a 15.0-m-tall building and throws a rock with a velocity of magnitude 30.0 m/s at an angle of 33.0° above the horizontal. You can ignore air resistance. Calculate the maximum height above the roof reached by the rock; **1**

   a. 12.6 m
   b. 11.7 m
   c. 13.6 m
   d. 14.2 m

6. Write an example of zero vector. **1**

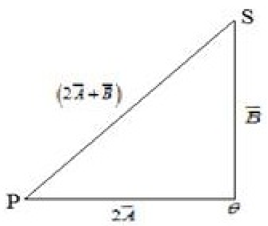
7. Two vectors of magnitude 3 units and 4 units are inclined at angle 60° w.r.t each other. Find the magnitude of their difference. **1**

8. The magnitude of vectors A, B and C are 12, 5 and 13 units respectively and A + B = C, find the angle between A and B. **1**

9. In dealing with the motion of a projectile in the air, we ignore the effect of air resistance on the motion. This gives trajectory as a parabola as you have studied. What would the trajectory look like if air resistance is included? sketch such a trajectory and explain why you have drawn it that way. **2**

10. An aircraft executes a horizontal loop of radius 1 km with a steady speed of 900 kmh^{-1}. Compare its centripetal acceleration with the acceleration due to gravity. **2**

11. A vector $\overline{A}$ has magnitude 2 and another vector $\overline{B}$ have magnitude 3 and is perpendicular to each other. By vector diagram find the magnitude of $2\overline{A} + \overline{B}$ and show its direction in the diagram. **2**

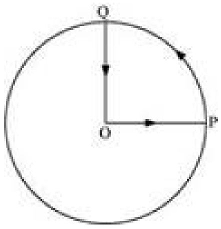


12. A projectile is projected with a certain velocity u at an angle θ with horizontal from the ground. Find expression for its trajectory. **3**

13. A projectile is fired with speed u making an angle θ with horizontal from the surface

of Earth. Prove that the projectile will hit the surface of earth with same speed and at the same angle. **3**

14. A cyclist starts from the centre O of a circular park of radius 1 km, reaches the edge P of the park, then cycles along the circumference, and returns to the centre along QO as shown in figure. If the round trip takes 10 min, what is the **3**



i. net displacement,
ii. average velocity, and
iii. average speed of the cyclist?

15. a. What is the angle between $\vec{A}$ and $\vec{B}$ if $\vec{A}$ and $\vec{B}$ denote the adjacent sides of a parallelogram drawn form a point and the area of the parallelogram is $\frac{1}{2}AB$?
    b. State and prove triangular law of vector addition. **5**